Name: B SREE NIKHIL REDDY ID number: 2520090058

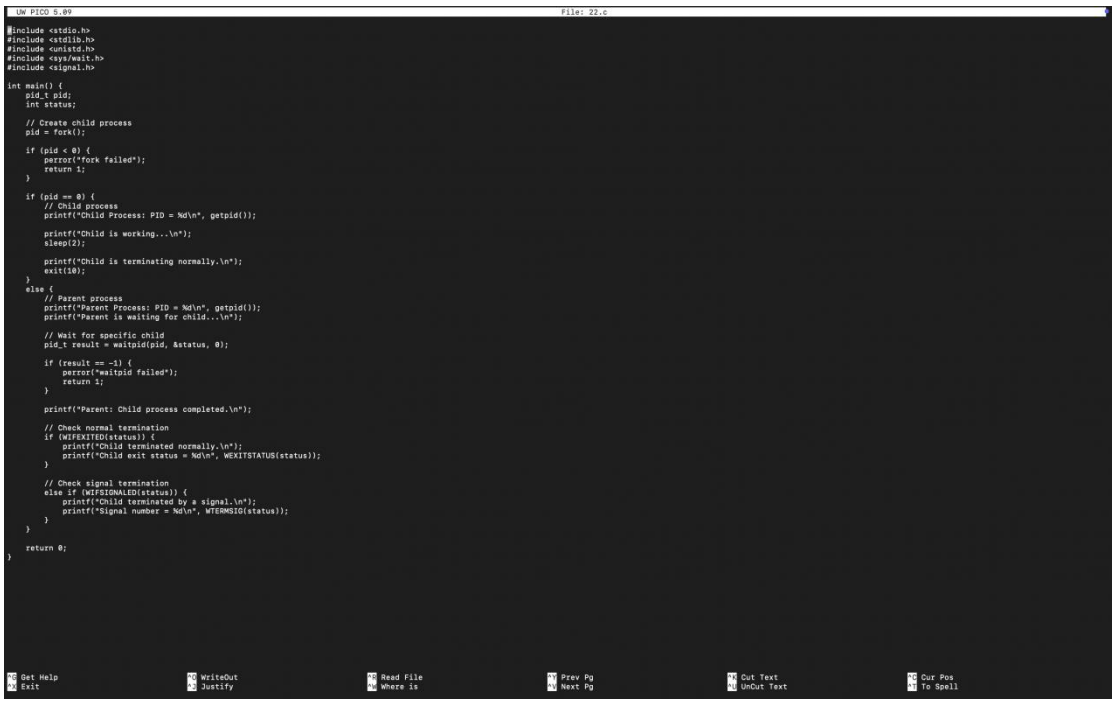
Week 4

SESSION-7

Task 1: Process Synchronization and Child Process Monitoring Using waitpid()

To create a child process, synchronize the parent with the child using waitpid(), monitor the child process, check normal termination, retrieve the exit status, handle signal termination, and test process synchronization.

CODE:



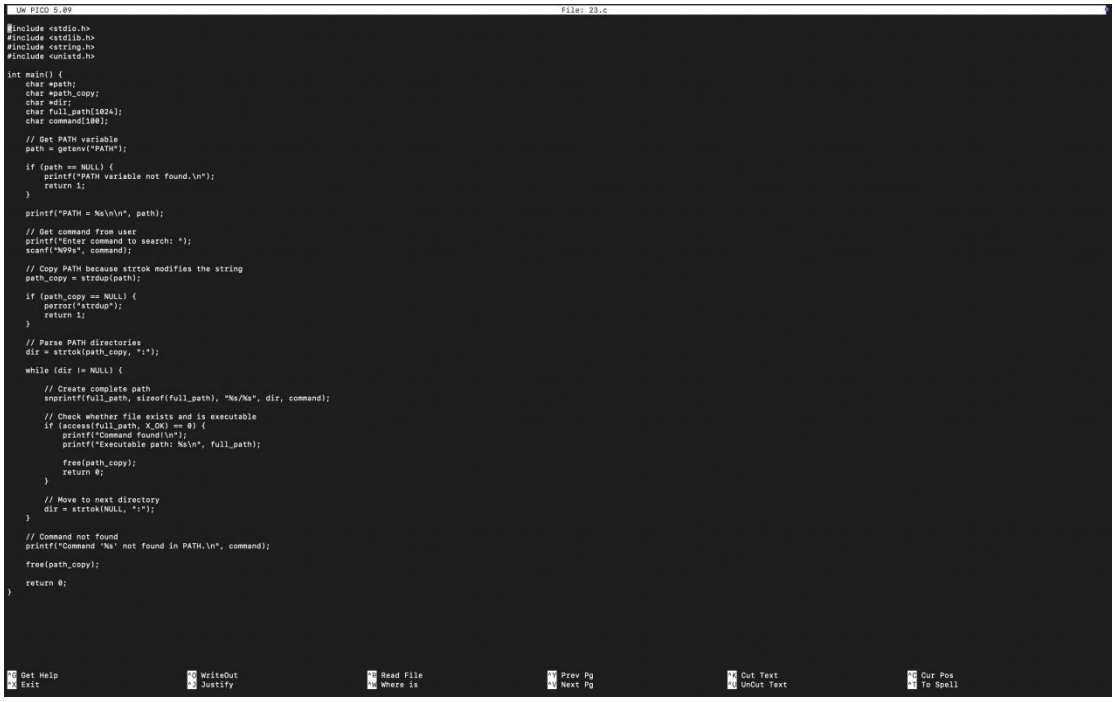
OUTPUT:



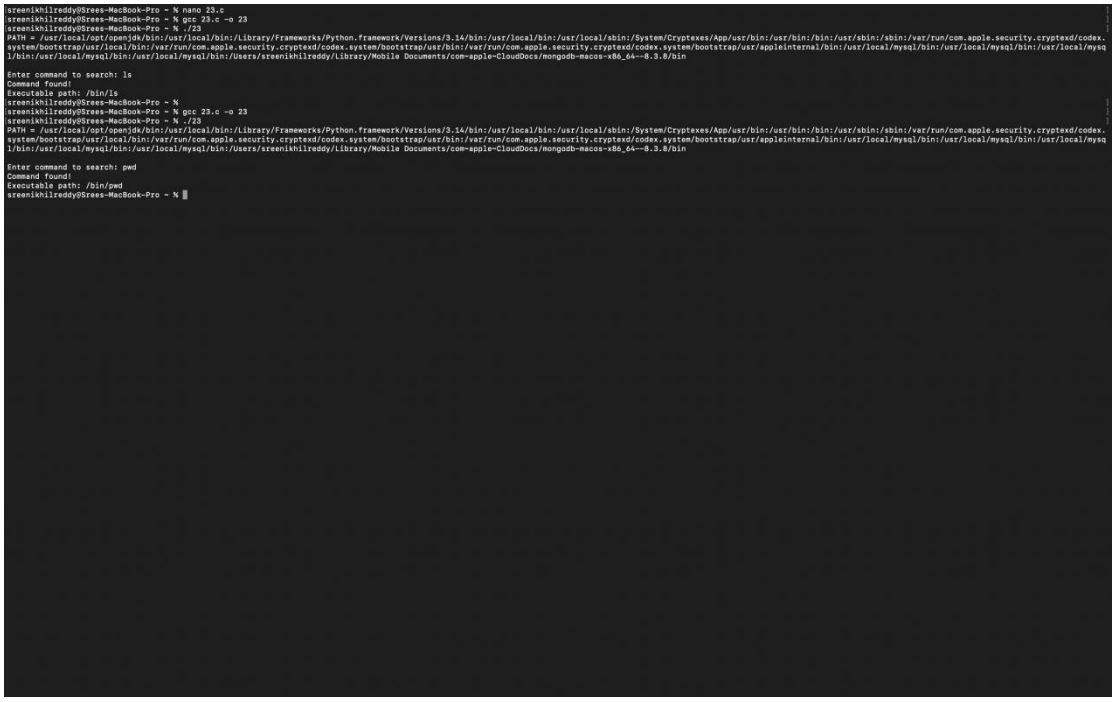
Task 2: PATH Variable Retrieval and Command Resolution

To retrieve the PATH variable, parse its search directories, locate executable files, verify execute permissions, handle missing commands, and test command resolution.

CODE:



OUTPUT:

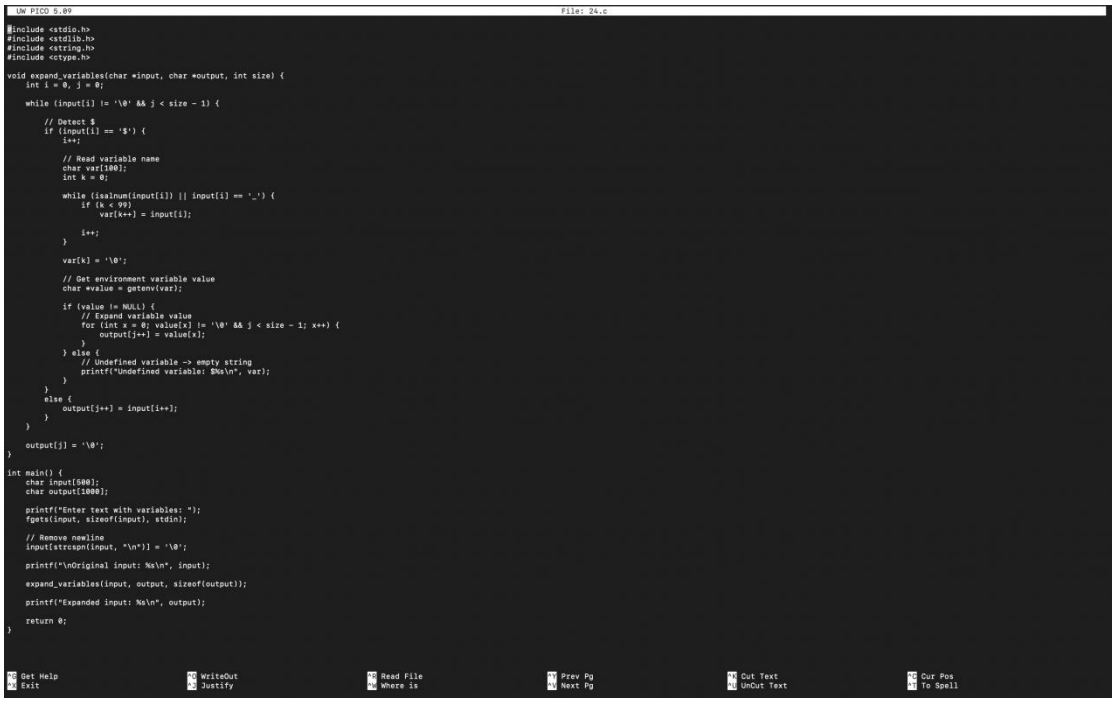


SESSION-8

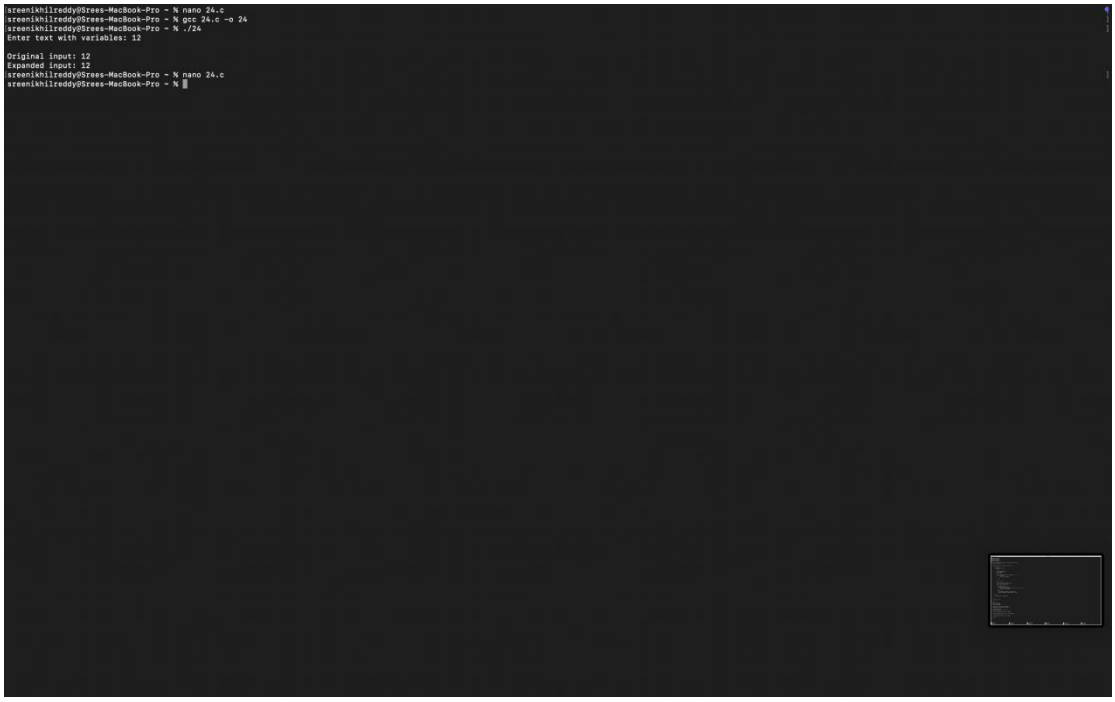
Task 1: Variable Reference Detection and Value Expansion

To detect variable references, expand environment variable values, handle undefined variables, update tokens after expansion, support nested variables, and test expansion logic.

CODE:



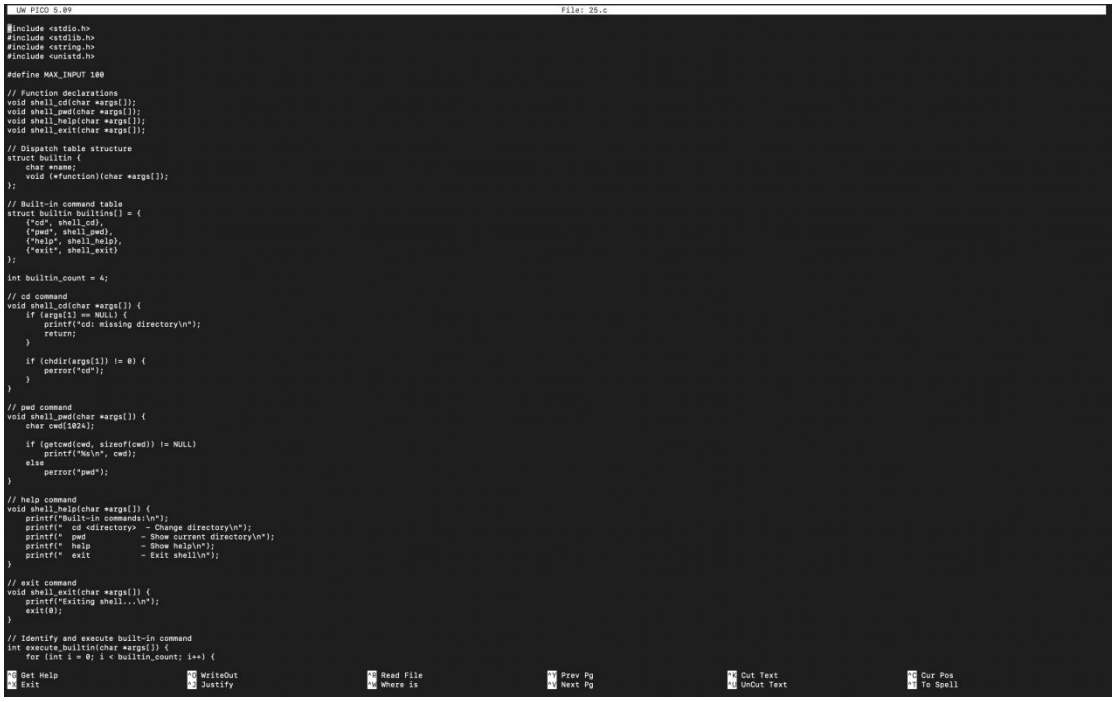
OUTPUT:



Task 2: Built-in Identification, Dispatch Table and In-Process Execution

To identify built-in commands, create a dispatch table, execute in-process commands, handle invalid commands, maintain shell state, and test dispatch logic.

CODE:



OUTPUT:


Last login: Thu Aug 28 12:14:05 on ttys000
sreenikhilreddy@Srees-MacBook-Pro ~ % nano 25.c
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc 25.c -o
clang: error: argument to '-o' is missing (expected 1 value)
sreenikhilreddy@Srees-MacBook-Pro ~ % gcc 25.c -o 25
sreenikhilreddy@Srees-MacBook-Pro ~ % ./25
Simple Shell
Type 'help' for commands.
myshell> exit
Exiting shell...
sreenikhilreddy@Srees-MacBook-Pro ~ % ./25
Simple Shell
Type 'help' for commands.
myshell> pws
Command not found: pws
myshell> pwd
/Users/sreenikhilreddy
myshell>